Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 101501
Subject: Analysis & Design of Concrete Structure
Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) IS 456 is permitted.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The center of gravity of a stable object should generally be
- (i) Very high
- (ii) Outside the base
- (iii) Low and within the base
- (iv) Constantly moving

(b) The strength of concrete achieved in structure is considered as
- (i) 0.67 x Cube strength of concrete
- (ii) 0.33 x Cube strength of concrete
- (iii) 0.17 x Cube strength of concrete
- (iv) 0.25 x Cube strength of concrete

(c) Limiting depth of neutral axis corresponds to
- (i) Balanced section
- (ii) Unbalanced section
- (iii) Over-reinforced section
- (iv) None of the above

(d) Minimum cement content for M20 mix in RCC is
- (i) 300 kg/m^3
- (ii) 375 kg/m^3
- (iii) 350 kg/m^3
- (iv) 280 kg/m^3

(e) In a slab, alternate bars are bent up at a distance
- (i) $\frac{l}{7}$ from centre of support
- (ii) $\frac{l}{5}$ from centre of support
- (iii) $\frac{l}{6}$ from centre of support
- (iv) None of these

(f) Intermediate beams in a slab floor system casts monolithically are called as
- (i) T-beams
- (ii) H-Beams
- (iii) L-Beams
- (iv) None of these

(g) Failure of bond in reinforced concrete may lead to
- (i) Increased compressive strength
- (ii) Slipping of reinforcement bars
- (iii) Proper load transfer
- (iv) Improved durability

(h) Isolated footings are designed for
- (i) Bending moment
- (ii) One-way shear
- (iii) Two-way shear
- (iv) All of the above

(i) The pitch of lateral ties should not be more than
- (i) 200 mm
- (ii) 250 mm
- (iii) 300 mm
- (iv) 100 mm

(j) Dog-legged stairs are designed similar to
- (i) One way slabs
- (ii) T-beams
- (iii) Two-way slabs
- (iv) None of theses

Q.2 (a) What are the role of engineer, architect, user and builder in building planning, design and construction? Explain briefly. **[7]**

(b) Discuss the various design philosophies for the design of concrete structure? State merits and demerits each. **[7]**

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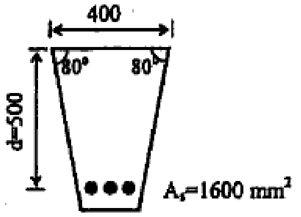
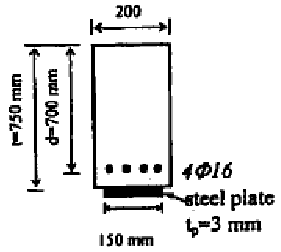
Q.3 (a) Find the ultimate moment capacity for the cross-section shown in Fig. Given $f_{ck} = 30$ N/mm², and $f_y = 415$ N/mm². **[7]**



(b) 3 mm steel plate with a yield strength of 400 N/mm² is glued to a concrete beam reinforced with steel bars (4ϕ16, f_y=360 N/mm²) as shown in figure. Determine the bending moment that the reinforced concrete section can resist. The concrete compressive strength of the beam is 20 N/mm². **[7]**



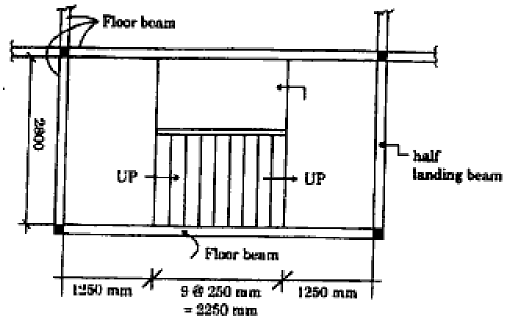
Q.4 (a) Design a doubly reinforced concrete cross-section to withstand an ultimate moment of 380 kNm knowing that the beam depth equals to 650 mm. width of beam = 250 mm, $f_{ck} = 35$ N/mm², and $f_y = 415$ N/mm². **[10]**

(b) Why helical ties are preferred over lateral links in circular column? **[4]**

Q.5 A simply supported RC slab, having a clear span of 3.2 m, is supported only on two opposite sides on brick walls of 250 mm thick. If the live load on the slab is 3 kN/m² and floor finish being 1.2 kN/m², design the RC slab as per limit state method. Use M30 concrete and Fe500 grade steel. Also check for deflection. Sketch the reinforcement details. Assume clear cover to the reinforcement as 20 mm. **[14]**

Q.6 A reinforced concrete short column with an effective height of 3 m carries an ultimate load of 1400 kN. The column has a rectangular cross-section measuring 300 mm × 600 mm and is reinforced with 8 longitudinal bars of 20 mm diameter. The column uses M25 grade concrete and Fe500 grade steel. The effective cover to the reinforcement is 60 mm. Determine the ultimate moment carrying capacity of the column about its major axis only. **[14]**

Q.7 A dog-legged staircase is to be designed for a building supported on beams and columns, as shown in the figure. The floor-to-floor height is 3 m. The beams, including the half-landing beams, have cross-sectional dimensions of 250 mm × 400 mm. The staircase is subjected to a live load of 3 kN/m². Using the limit state method and assuming M25 grade concrete and Fe500 grade steel, design and detail the reinforcement for a typical flight of the staircase with suitable sketches. The waist slab thickness may be taken as 150 mm. Deflection check is not required. **[14]**



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Q.8 (a) Design a strap footing for two columns (C_1 and C_2) 400 mm x 400 mm and 600 mm x 600 mm in size spaced at 4.5 m centre to centre distance. Columns C_1 is carrying a factored load of 900 kN and C_2 is carrying a factored load of 1500 kN. Column C_1 is 400 mm away from the property line. The safe bearing capacity of soil is 150 kN/m^2 [7]

(b) An RCC beam 250 mm x 250 mm has a clear span of 5.5 m. the beam has 2-20 mm ϕ bars going into the support. Factored shear force is 140 kN. Check for development length if Fe 415 and M20 grade of concrete is used. [7]

Q.9 Write short note on ***any four*** of the following: **[3.5 x 4 = 14]**

(a) Limit state of serviceability
(b) Effect of Wind and Earthquake loads on RCC structures
(c) First principles of process of design
(d) Principles of stability
(e) Under-reinforced and over-reinforced beams
(f) Functions of various types of transverse reinforcement in RCC structure

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